

RealRF-DG: Source-Aware Dual-Domain Learning for Data-Efficient Radio Fingerprinting Across Real Capture Environments

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Abstract

Radio-frequency fingerprinting models are frequently evaluated on generated signals or a single capture setup, making it difficult to distinguish device features from dataset shortcuts. We present RealRF-DG, a reproducible multi-source benchmark and dual-domain model trained on deterministic 10% subsets of three real-signal corpora. The model fuses temporal raw-I/Q features with spectral magnitude and phase-step features, uses source-specific device heads, and applies gradient-reversal regularization to reduce source leakage. All dataset selections, splits, hardware tuning decisions, and predictions are exported as auditable artifacts.

CCS Concepts

• Computing methodologies → Neural networks.

Keywords

radio-frequency fingerprinting, domain generalization, deep learning, real signals, physical-layer authentication

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1 Introduction

State the practical authentication problem, the risk of generated-signal benchmarks, and the paper contributions.

2 Related Work

Cover RF fingerprinting, cross-domain robustness, complex-signal neural networks, and data-efficient learning.

3 Datasets and Protocol

We use only publisher-released real captures from WIDEFT, INRIA PLA, and the WLAN environment corpus [1–3].

Table 1: Replace placeholders only with completed experiment results.

Source	Accuracy	Macro F1	ECE
WIDEFT	–	–	–
INRIA PLA	–	–	–
WLAN office holdout	–	–	–

4 Method

5 Experimental Setup

Report all hyperparameters, the measured L4 batch size, software versions, three random seeds, and the ablation plan.

6 Results

7 Limitations and Ethics

Discuss receiver/channel confounding, the 10% compute constraint, lawful use, device tracking risks, and limits on generalization.

8 Conclusion

Summarize findings without overstating deployment readiness.

References

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Stage A: Real-Signal Curation and Reproducible 10% Sampling

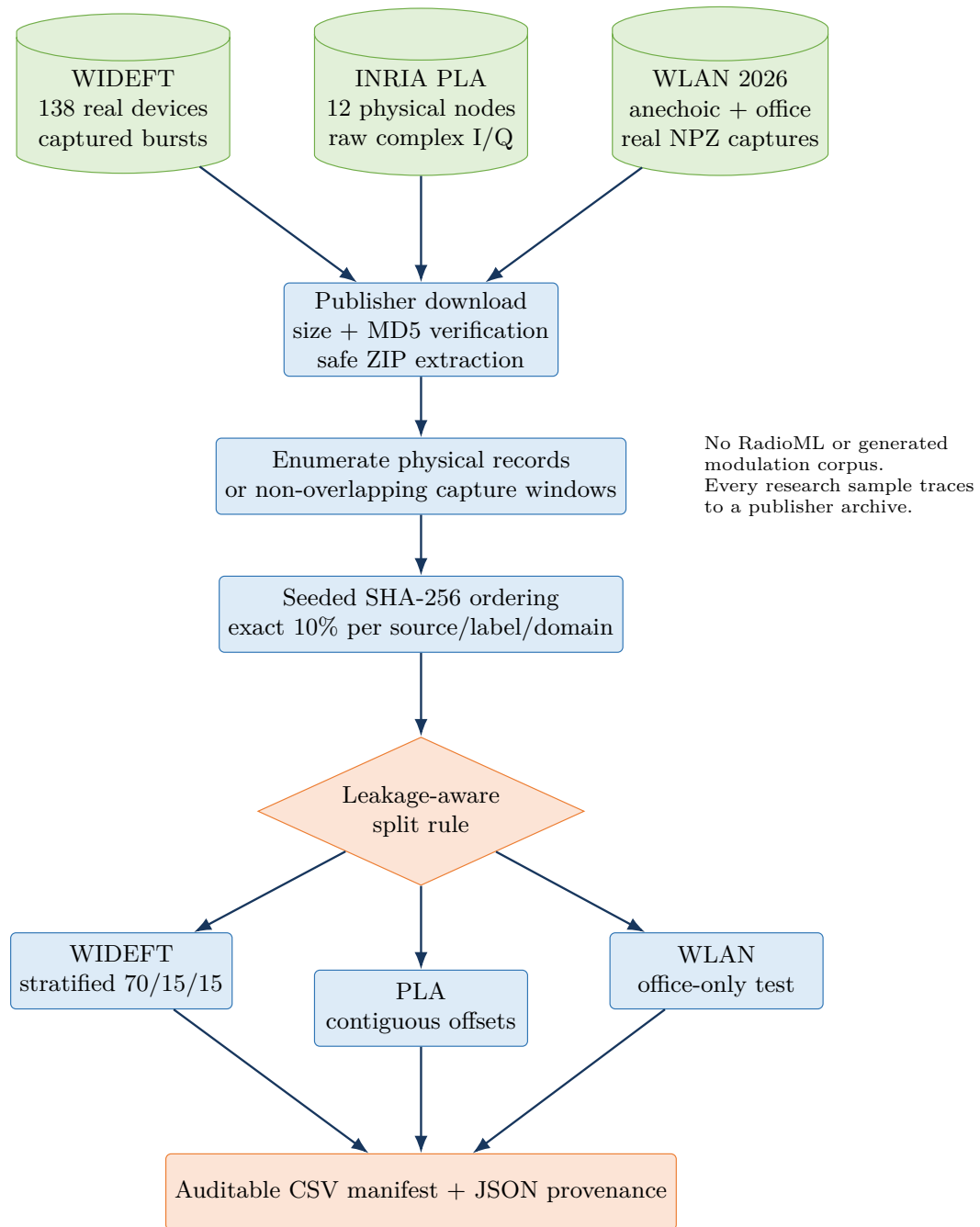


Figure 1: Checksum-verified real-data curation and deterministic 10% sampling.

Stage B: Dual-Domain Source-Aware RF Encoder

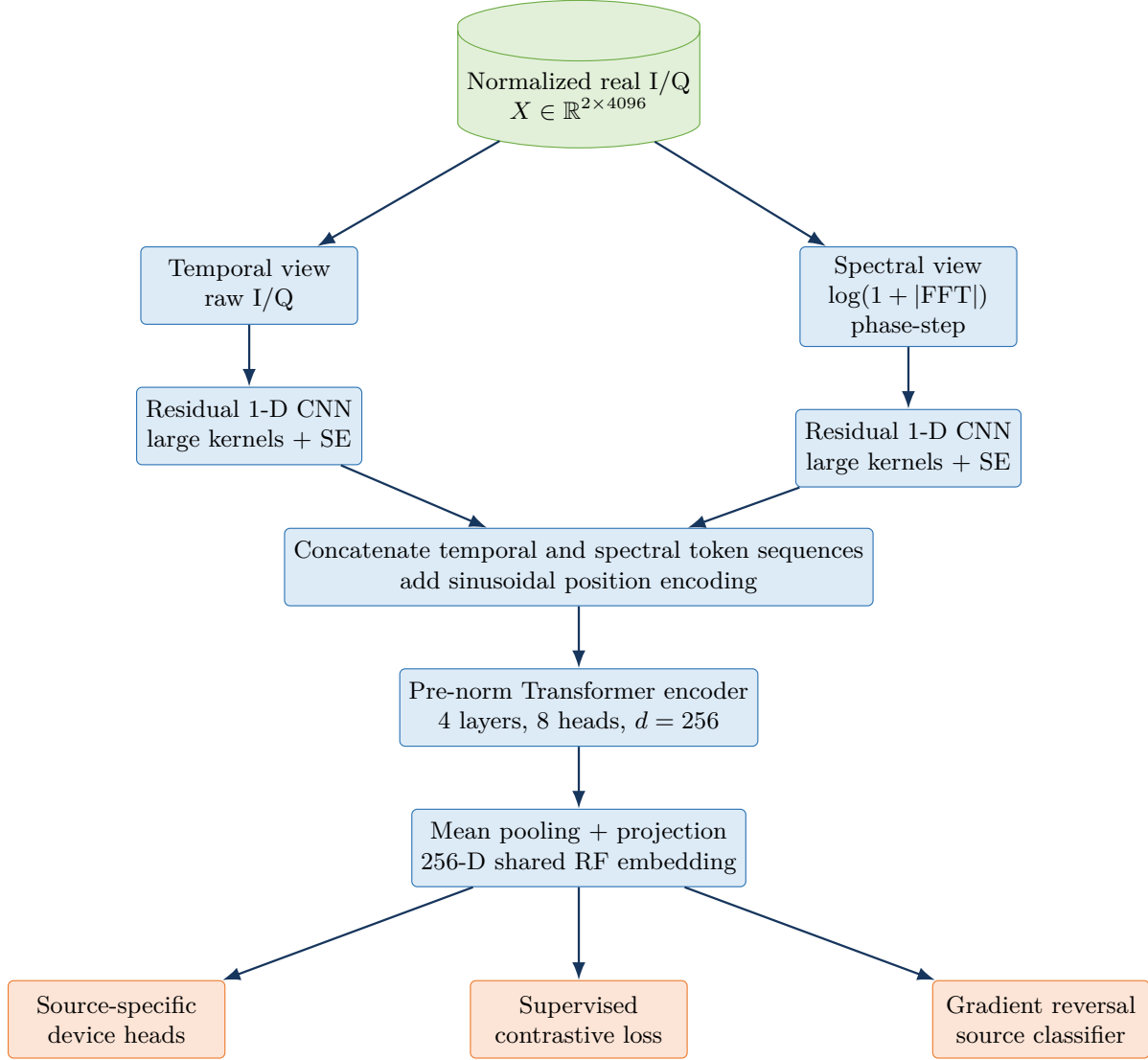


Figure 2: Dual-domain source-aware RF fingerprinting architecture.